2019春高数下期末试题(回忆版)

1. Determine whether the following statements are true or false? No justification is necessary.

(1) If $a_n>0, \forall n$, and $\lim\limits_{n\to\infty} n a_n=0$, then $\sum\limits_{n=1}^{\infty} a_n$ converges.

(2) The plane $x+y-2z=1$ is perpendicular to the plane $x+y+z=1$.

(3) If $f(x,y)$ has two local maxima, then f must have a local minimum.

2. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Which one of the following series diverges?

(A) $\sum\limits_{n=1}^{\infty} \frac{n^2}{2^n}$. (B) $\sum\limits_{n=1}^{\infty} \frac{1}{n^{1+\frac{1}{n}}}$. (C) $\sum\limits_{n=1}^{\infty} \frac{(-1)^n \ln n}{\sqrt{n}}$ (D) $\sum\limits_{n=1}^{\infty} \frac{(-1)^n (3+(-1)^n\cdot 2)^n}{6^n}$

(2) The iterated integal $\int_0^{\frac{\pi}{2}} \int_0^{\cos\theta} f(r\cos\theta, r\sin\theta)\, r\, dr d\theta$ can be written as

(A) $\int_0^1 \int_0^{\sqrt{y-y^2}} f(x,y)\, dxdy$. (B) $\int_0^1 \int_0^{\sqrt{1-y^2}} f(x,y)\, dxdy$.

(C) $\int_0^1 \int_0^1 f(x,y)\, dydx$. (D) $\int_0^1 \int_0^{\sqrt{x-x^2}} f(x,y)\, dydx$.

(3) For the function, $f(x,y)=\begin{cases} \frac{2xy}{\sqrt{x^2+y^2}}, & (x,y)\neq(0,0) \\ 0, & (x,y)=(0,0) \end{cases}$; which of the following statements is correct?

(A) f is not continuous at $(0,0)$.

(B) f is continuous at $(0,0)$, but its partial derivative f_x and f_y do not exist at $(0,0)$.

(C) Both partial derivatives f_x and f_y exist everywhere and are also continuous at $(0,0)$.

(D) f is not differentiable at $(0,0)$.

(4) For the critical points of the function $f(x,y)=2x^4+y^4-2x^2-2y^2$, which one of the following statements is correct?

(A) $(0,0)$ is a local minima. (B) $(0,1)$ is a local maxima.

(C) $(0,-1)$ is a saddle point. (D) There are no local maxima among all the critical points.

(5) If the function $f(x,y)$ has the continuous first partial derivatives $\frac{\partial f}{\partial x}>0$ and $\frac{\partial f}{\partial y}<0$, $\forall (x,y)\in R^2$, which one of the following statements is correct?

(A) $f(0,0)>f(1,1)$ (B) $f(0,0)<f(1,1)$ (C) $f(0,1)>f(1,0)$ (D) $f(0,1)<f(1,0)$

3. Please fill in the blank for the questions below.

(1) Compute the limit $=\lim\limits_{(x,y)\to(0,0)} \frac{\sqrt{x^2+y^2+1}-1}{x^2+y^2}=$ _____.

(2) The direction (unit vector) in which the function $f(x,y)=x^2+xy+y^2-y$ increases most rapidly at the point $(-1,+2)$ is _____.

(3) $\int_0^1 \int_y^1 \frac{\tan x}{x}\, dxdy=$ _____.

4. (1) Find the interval of convergence of the series $\sum\limits_{n=2}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{\sqrt{n+9012}\ln n}$.

(2) For what values of x does the series converge absolutely, or conditionally?

5. The region D is bounded by $z=\sqrt{x^2+y^2}$ and $z=\sqrt{1-x^2-y^2}$. Consider the following integral

$$\iiint_D (x+z)\, dxdydz,$$

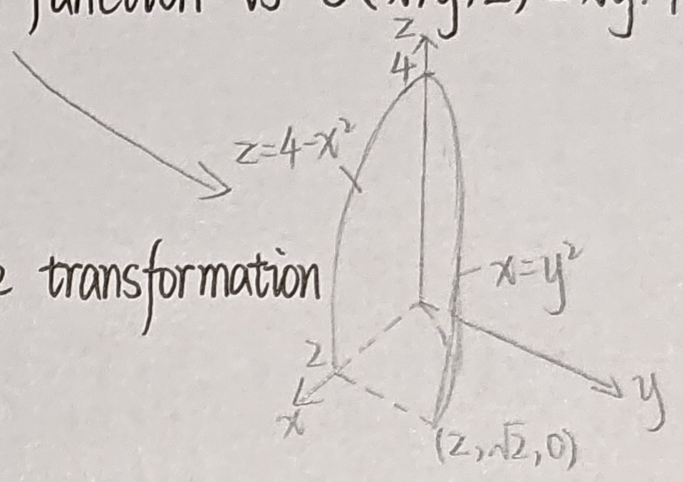
(1) Convert the above integral to an equivalent iterated integral in cylindrical coordinates;

(2) Convert the above integral to an equivalent iterated integral in spherical coordinates.

6. Assume we can put a cuboid into the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$. Use the mothed of Lagrange multipliers to find the length, width and height of the cuboid such that it achieve the maximum volume.

7. Find the equation of the osculating circle for the parabola $y=x^2$ at $x=1$.

8. A solid in the first octant is bounded by the planes $y=0$ and $z=0$ and by the surfaces $z=4-x^2$ and $x=y^2$ (see the figure below). Its density function is $\delta(x,y,z)=xy$. Find the center of the mass for the solid.



9. Use the substitution in double integal (please find the transformation by yourself) to evaluate the integral $\iint_D e^{\frac{y-x}{y+x}}\,dxdy$,

here D is the triangular region bounded by the lines $x=0$, $y=0$, and $x+y=2$.

10. Consider the line integral $\int_{(1,1,1)}^{(1,3,\pi)} e^x \ln y\,dx+\left(\frac{e^x}{y}+\sin z\right)dy+y\cos z\,dz$.

(1) Show that the differential form in the integral is exact.

(2) Evaluate the integral.

11. Evaluate $\iint_S \nabla\times(4x\mathbf{j})\cdot n\,d\sigma$,

where S is the hemisphere $x^2+y^2+z^2=16$, $z\geq 0$. Use the normal vectors pointed away from the origin.

12. Find the outward flux of $F=(6x+y)\mathbf{i}-(x+z)\mathbf{j}+4yz\,\mathbf{k}$ across the boundary of D, where D is the region in the first ~~about~~ octant bounded by the cone $z=\sqrt{x^2+y^2}$, the cylinder $x^2+y^2=1$, and the coordinate planes.

13. The sequences $\{a_n\}$ and $\{b_n\}$ satisfy $0<a_n<\frac{\pi}{2}$, $0<b_n<\frac{\pi}{2}$, and $\cos a_n - a_n = \cos b_n$, $n=1,2,3,\cdots$. The series $\sum_{n=1}^{\infty} b_n$ converges. Show that $\lim_{n\to\infty} a_n = 0$.